

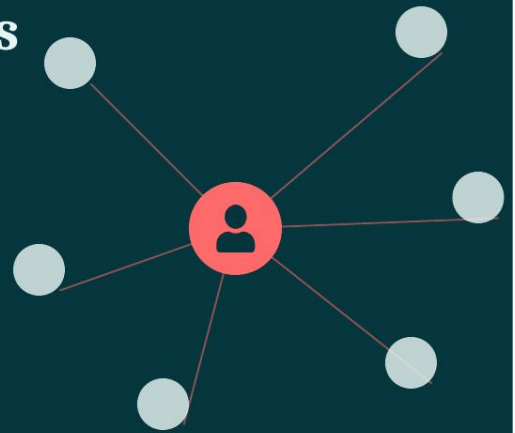
E-COMMERCE · A BEGINNER'S GUIDE

How Recommendation Systems Work

A hands-on, jargon-light introduction

For aspiring data scientists

Presented by Your Name · June 2026



Speaker notes

Welcome the learners. This deck builds intuition from the ground up with small worked examples rather than heavy math. By the end they should understand the two core approaches, the data behind them, how systems are evaluated, and how to build a first baseline. Encourage questions throughout — it's a teaching session, not a briefing.

MODIFIABLE FOR YOUR PLATFORM

- Swap in your name and the session date.

WHAT YOU'LL LEARN

Five things you'll be able to explain by the end



The problem

Why recommendations exist and what they solve



Content-based filtering

Recommending by item features



Collaborative filtering

Recommending from behavior patterns



Evaluation

How we measure if recommendations are good



Getting started

Tools and datasets to build your first model

Speaker notes

Set expectations as learning objectives so people know where they're headed. Reassure them that the math stays light — we lean on small concrete examples. Revisit this list at the end to confirm each objective was met.

THE PROBLEM, INTUITIVELY

Imagine a store with a billion shelves



You can't browse it all

Nobody can scroll a billion products. Most never get seen.



Too many choices freeze us

Faced with endless options, people often buy nothing.



A good guide helps

A recommender is like a clerk who remembers everyone's taste.

Why it pays off

17%

of global online orders driven by AI recommendations & personalization

+7%

higher average order value for retailers using generative AI

Salesforce Shopping Index, 2024.

Speaker notes

Lead with the analogy — a store with a billion shelves — because it makes the abstract concrete. Connect the two human problems (can't browse it all, too many choices paralyze us) to why a recommender is valuable: it's an automated, personalized guide. The stats are recent (Salesforce's 2024 Shopping Index, 1.5 billion shoppers): AI-driven recommendations and personalization drove 17% of global online orders, and retailers using generative AI saw 7% higher average order value.

MODIFIABLE FOR YOUR PLATFORM

- These are recent industry figures (Salesforce, 2024); swap in your own platform's data if you have it.

ABBREVIATIONS

- AOV = Average Order Value
- AI = Artificial Intelligence

SOURCES

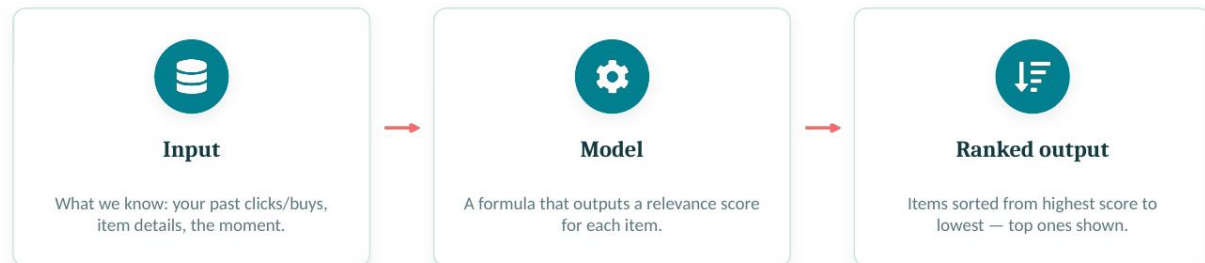
- Salesforce Shopping Index, 2024 — AI drove 17% of global orders; +7% AOV with generative AI (analysis of 1.5B shoppers).

THE CORE IDEA

Predict a score, then sort by score



Almost every recommender does the same thing: give each item a “how much will this user like it?” score, then show the highest-scoring items first.



Speaker notes

Demystify the whole field in one sentence: score every item, sort, show the top. Everything else — content-based, collaborative, deep learning — is just a different way of computing that score. Keep returning to this mental model as the methods get more sophisticated.

APPROACH 1 — CONTENT-BASED FILTERING

“You liked this, so here's something similar”

Recommend items whose features match what the user already liked.

Example

A shopper liked: “Trail Running Shoe”

Tags: sport · outdoor · lightweight · rubber sole

So we look for items sharing those tags:

- ✓ **Hiking Sandals** outdoor · lightweight
- ✓ **Gym Trainers** sport · rubber sole
- ✗ **Office Loafer** formal · leather

Good to know

- + Works for brand-new items (we know their features)
- + Easy to explain: “because it's also outdoor gear”
- Can feel repetitive — stays close to what you know
- Needs good item data (tags, descriptions)

Speaker notes

Walk the example slowly. Content-based filtering compares item features (tags, categories, text). The shopper liked a trail running shoe, so items sharing tags like 'outdoor' or 'lightweight' score higher; an office loafer shares none, so it scores low. Emphasize the trade-off: great for new items and easy to explain, but it can trap users in a narrow lane and depends on good item metadata.

MODIFIABLE FOR YOUR PLATFORM

- Replace the shoe example with a product from your own catalog to make it relatable to your team.

APPROACH 2 — COLLABORATIVE FILTERING

“People like you also liked...”

Ignore item features. Learn from behavior: find users with similar taste, then recommend what they liked.

	Item A	Item B	Item C	Item D
You	★★★★	★★	—	?
Ana	★★★★	★★	—	★★★★
Ben	★	★★★★	★★★	★
Cara	★★★★	★★	★	★★★★

Reading the table

- 1 You and Ana rate Items A & B almost identically.
- 2 Ana loves Item D (★★★★). You haven't tried it.
- 3 So we predict you'll like Item D too → recommend it.
- 4 That's “user-based” CF. “Item-based” flips it: find items rated alike.

Speaker notes

This is the heart of the teaching deck — work the matrix live. Point out that we never used any item features; we only used ratings (behavior). ‘You’ and Ana agree on Items A and B, so they’re similar users. Ana rated Item D four stars and you haven’t seen it, so the system predicts you’ll like Item D and recommends it (the ‘?’ cell). Then explain the two flavors: user-based (find similar users) vs item-based (find items rated similarly), and note item-based tends to scale better — that’s what Amazon famously used.

ABBREVIATIONS

- CF = Collaborative Filtering

FROM BIG TABLES TO EMBEDDINGS

Turn taste into coordinates; nearby = similar

Real rating tables are huge and mostly empty (most people rate almost nothing).

Embeddings compress each user and item into a short list of numbers — a point in space.

Items close together are similar; we recommend items near what you like.



Speaker notes

Bridge from the concrete table to the idea powering modern systems. The rating matrix is enormous and sparse; embeddings squeeze each user/item into a handful of numbers — coordinates in a 'taste space.' In the picture, sporty products cluster together and formal products elsewhere; recommending becomes 'find nearby points.' This is the intuition behind matrix factorization and the graph embeddings Alibaba's Taobao uses at billion-item scale. Keep it visual; no equations needed.

ABBREVIATIONS

- Embedding = a compact list of numbers (a vector) representing a user or item

THE DATA BEHIND IT

What users say vs. what they do



Explicit signals

Users tell us directly

- ✓ Star ratings you give
- ✓ Reviews you write
- ✓ Likes & wishlists
- ✓ Preferences you pick



Implicit signals

We watch behavior

- ✓ Clicks and taps
- ✓ Things you buy
- ✓ How long you look
- ✓ Items added to cart



Explicit data is clear but rare (most people never rate). Implicit data is everywhere but noisy — a click isn't always a “like.”

Speaker notes

Distinguish the two data types and stress the practical reality: you'll mostly work with implicit signals because they're abundant, but they're noisy — a click might be a misclick, a long dwell might mean confusion, not interest. Beginners often over-trust implicit signals; flag that early.

THE COLD-START PROBLEM

What to do when there's no history yet



New user

Someone just signed up — we don't know their taste yet.



New item

A product nobody has interacted with — nothing to learn from.

Common fixes



Show popular items to everyone at first



Ask a quick "what are you into?" at sign-up



Recommend new items by their features (content-based)



Use clues: location, device, time of day

Speaker notes

Cold start is a classic chicken-and-egg problem and a favorite interview topic. New users have no history; new items have no interactions. The fixes are practical: fall back to popularity, ask onboarding questions to seed taste, use content features to place new items, and lean on context. Note that content-based filtering is the natural rescue for new items because it doesn't need interactions.

HOW DO WE KNOW IT'S GOOD?

Two metrics, explained with a tiny example

We recommended 3 items. Did the user like them?

✓	Item A	liked
✓	Item B	liked
✗	Item C	ignored

Precision@3 = 2 of 3 = 0.67

NDCG also rewards putting the liked items higher up the list.

Two kinds of testing



Offline (lab)

Test on past data. Cheap and fast. Metrics: precision@k, recall, NDCG.



Online (real users)

Run an A/B test on live traffic. Metrics: click-through, conversion.

Speaker notes

Make the metrics tangible. Precision@k just asks: of the k items we showed, how many were relevant? In the example, 2 of 3 liked = 0.67. Recall asks how many of all the good items we managed to surface. NDCG additionally rewards ranking the best items highest. Then the crucial distinction: offline metrics are cheap proxies on past data, but only an online A/B test on real users tells you it actually works. Beginners often stop at offline — warn against it.

ABBREVIATIONS

- Precision@k = fraction of the top-k recommended items that are relevant
- Recall = fraction of all relevant items that were successfully recommended
- NDCG = Normalized Discounted Cumulative Gain (ranking-quality score that rewards relevant items near the top)
- A/B test = randomized live experiment comparing two versions

REAL SYSTEMS IN THE WILD

The same ideas, at enormous scale



Amazon

Item-based collaborative filtering — the “bought this also bought” idea you just learned.



Taobao / Alibaba

Graph embeddings place a billion items in “taste space” to power the “Guess What You Like” feed.



JD.com (Jingdong)

Deep-learning models match shoppers to products across a massive catalog in milliseconds.

Speaker notes

Connect the lesson to reality so it doesn't feel academic. Amazon is the textbook item-based CF example you just saw in the matrix. Taobao (Alibaba) builds an item graph from behavior and learns embeddings — exactly the 'taste space' idea — to feed its 'Guess What You Like' homepage across a billion items. JD.com uses deep learning to match shoppers to products in milliseconds. Same core concepts, just scaled up massively.

ABBREVIATIONS

- CF = Collaborative Filtering
- JD = Jingdong (JD.com)

BEGINNER PITFALLS & GLOSSARY

Mistakes to dodge, terms to know

Common pitfalls

- ⚠️ Trusting offline metrics only — always A/B test
- ⚠️ Data leakage: testing on data the model already saw
- ⚠️ Recommending only popular items (popularity bias)
- ⚠️ Ignoring cold start for new users and items

Quick glossary

- CF** — Collaborative Filtering
- AOV** — Average Order Value
- CTR** — Click-Through Rate
- NDCG** — ranking quality, top-weighted
- Embedding** — item/user as a vector

Speaker notes

Two practical lists. The pitfalls are the mistakes beginners actually make — especially trusting offline metrics and accidental data leakage (training and testing on overlapping data). The glossary collects the abbreviations used throughout so learners have a single reference. Encourage them to screenshot this slide.

ABBREVIATIONS

- CF = Collaborative Filtering
- AOV = Average Order Value
- CTR = Click-Through Rate
- NDCG = Normalized Discounted Cumulative Gain

YOUR NEXT STEPS

Build your first recommender this week



Get a dataset

Start with MovieLens — small, clean, and classic for learning.



Pick a library

Try Surprise, LightFM, or implicit for ready-made models.



Build a baseline

Start with popularity, then item-based collaborative filtering.



Evaluate it

Split train/test, measure precision@k, then iterate.

Tip: get a simple baseline working end-to-end first — then improve one piece at a time.

Speaker notes

End with momentum: a concrete starter path. MovieLens is the standard learning dataset; Surprise/LightFM/implicit are friendly libraries. The key habit to instill: get a tiny baseline running end-to-end before optimizing — a working popularity model beats a half-built neural network. Then improve one component at a time and measure each change.

MODIFIABLE FOR YOUR PLATFORM

- Swap in your team's preferred stack or an internal dataset if you have one available for practice.

KEY TAKEAWAYS

What you can now explain

- 1 Recommenders score every item, then show the highest-scoring ones
- 2 Content-based = match item features; collaborative = learn from behavior
- 3 Embeddings turn taste into coordinates; nearby items are similar
- 4 Implicit data is plentiful but noisy; watch out for cold start
- 5 Measure offline (precision@k, NDCG) but decide with online A/B tests

Now you have the map — go build your first recommender!

Thank you

Speaker notes

Revisit the learning objectives from slide 2 and confirm each is covered. Encourage learners to take the next-steps slide and actually build something this week — hands-on practice cements all of this far better than more slides.

ABBREVIATIONS

- NDCG = Normalized Discounted Cumulative Gain
- A/B test = randomized live experiment

BONUS SECTION

How real systems grew up

A friendly tour of how Amazon, Taobao, and JD.com built their recommenders — from the classics to today's AI — and who benefits most. Every number comes from 2024–2026 sources.

Speaker notes

Appendix divider. These slides support the deck with each platform's documented recommender history — including the post-GPT (generative/LLM) era — and effectiveness by user class. Every quantitative figure is from a source published after 2024.

BONUS · HOW REAL SYSTEMS GREW UP

Amazon — recommendation system history



How Amazon's recommendations grew from "also bought" lists into a chat assistant.



Sources: Amazon Science (2024); Amazon Q4 2025 earnings; COSMO — ACM SIGMOD 2024; Linden et al., IEEE Internet Computing (2003, historical).

Speaker notes

Teach the arc simply: it started as "people who bought this also bought" (1998), Amazon published how it worked (2003), then pages got personalized (2017). The big recent change is Rufus (2024) — a chat assistant powered by large language models — which by 2025 had 300M+ users and drove ~\$12B extra sales. The lesson: recommendations went from matching past purchases to understanding what you mean.

ABBREVIATIONS

- CF = Collaborative Filtering
- RAG = Retrieval-Augmented Generation
- GenAI = Generative AI

SOURCES

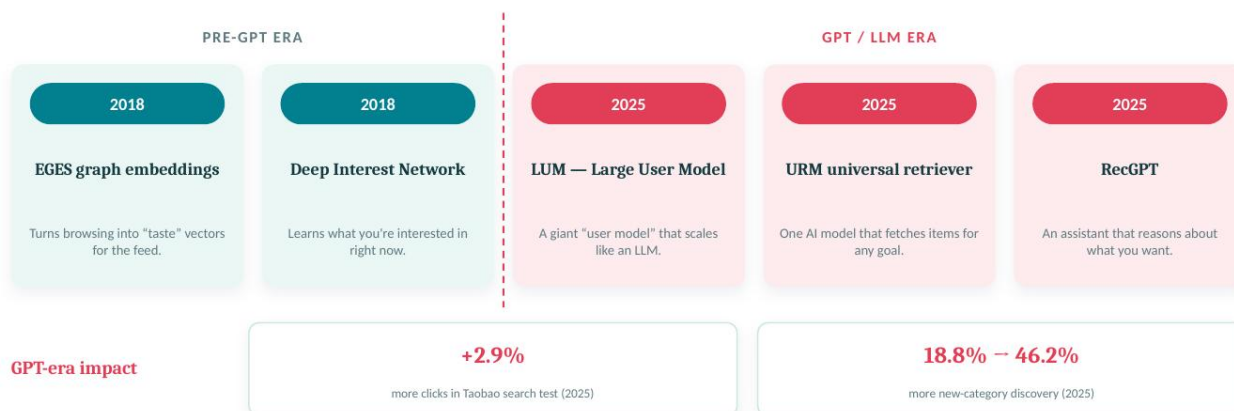
- Amazon Science (2024); Amazon Q4 2025 earnings; COSMO, ACM SIGMOD 2024; Linden et al., IEEE (2003).

BONUS · HOW REAL SYSTEMS GREW UP

Taobao / Alibaba — recommendation system history



How Taobao went from “taste vectors” to AI that reasons about what you want.



Sources: Wang et al., KDD 2018 (EGES); Zhou et al., KDD 2018 (DIN); Yan et al., 2025 (LUM); Jiang et al., 2025 (URM, arXiv:2502.03041); Yi et al., 2025 (RecGPT).

Speaker notes

Taobao first turned browsing into “taste vectors” (2018). Recently (2025) it built LLM-based systems: LUM (a giant user model), URM (one model that fetches items for any goal), and RecGPT (an assistant that reasons about intent). Concrete wins: +2.9% more clicks, and far more discovery of new categories (18.8%→46.2%).

ABBREVIATIONS

- EGES = Enhanced Graph Embedding w/ Side info
- DIN = Deep Interest Network
- CTR = Click-Through Rate
- RPM = Revenue Per Mille
- LUM = Large User Model
- URM = Universal Retrieval Model

SOURCES

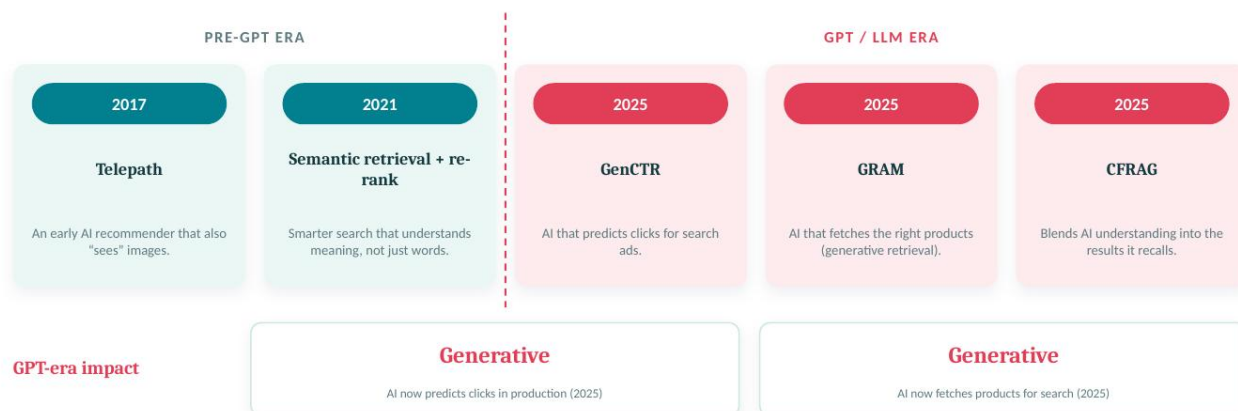
- Yan et al., LUM (2025); Jiang et al., URM (2025, arXiv:2502.03041); Yi et al., RecGPT (2025); Wang et al., EGES (KDD 2018).

BONUS · HOW REAL SYSTEMS GREW UP

JD.com (Jingdong) — recommendation system history



How JD.com moved from early AI recommenders to generative AI in production.



Sources: Zhao et al. (Telepath), 2017; Li et al., 2021; Kong et al., 2025 (GenCTR, arXiv:2507.11246); JD GRAM & CFRAG, 2025.

Speaker notes

JD's path: an early AI recommender that also looked at images (2017), smarter meaning-based search (2021), and now generative AI that predicts clicks and fetches products (2025). JD shares fewer public numbers, so we describe the direction rather than exact figures.

ABBREVIATIONS

- CTR = Click-Through Rate
- RAG = Retrieval-Augmented Generation
- GRAM = Generative Retrieval and Alignment Model

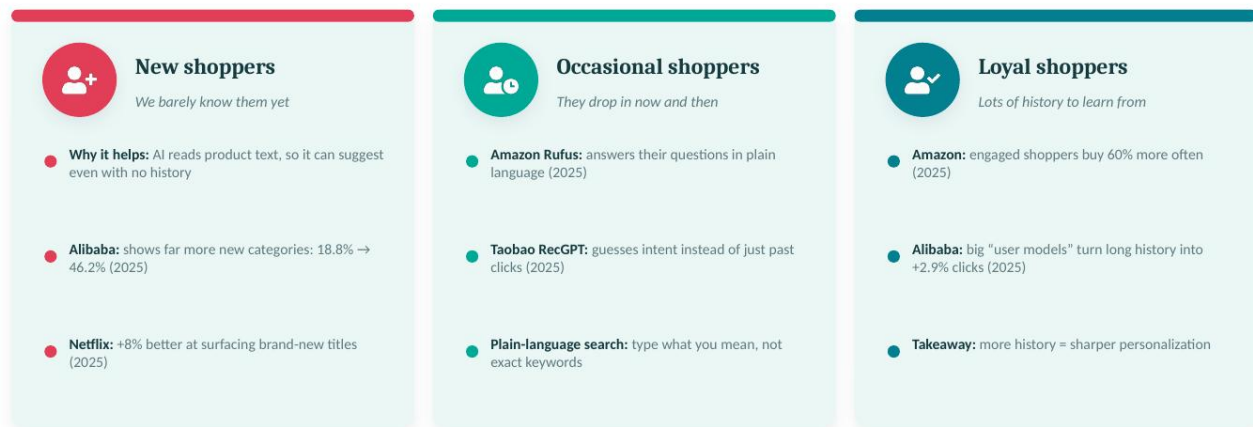
SOURCES

- Kong et al., GenCTR (2025, arXiv:2507.11246); JD GRAM & CFRAG (2025); Zhao et al. (2017); Li et al. (2021).

BONUS · HOW REAL SYSTEMS GREW UP

Who does AI help most? (by shopper type)

Different shoppers get different benefits. The big idea: AI helps new users most, because it can read words — not just past clicks.



Sources (2025): Jiang et al. (URM); Yan et al. (LUM); Amazon Q4 2025; Li et al., Nov 2025 (Netflix); cold-start video study, Aug 2025. Platform-level per-segment figures are rarely disclosed; cross-industry studies used as directional evidence.

Speaker notes

Plain lesson for learners: recommendation quality isn't the same for everyone. New users are the hardest (the "cold-start" problem) — and this is exactly where LLM-based systems help most, because they can read product descriptions and language instead of relying only on past clicks. Loyal users with lots of history were already well served; they get incremental gains. Note the honesty caveat: platforms seldom publish exact per-group numbers, so these are drawn from 2025 studies.

ABBREVIATIONS

- CF = Collaborative Filtering
- Recall@k = share of relevant items in top k
- CTR = Click-Through Rate
- RAG = Retrieval-Augmented Generation
- URM = Universal Retrieval Model
- LUM = Large User Model

SOURCES

- Jiang et al., URM (2025); Yan et al., LUM (2025); Amazon Q4 2025; Li et al., Nov 2025 (Netflix cold-start); cold-start video study, Aug 2025.

APPENDIX · SOURCES

Where these facts come from (2024–2026)

Amazon — Rufus	Amazon Science, “The technology behind Amazon’s GenAI shopping assistant, Rufus” (2024); scale & sales via Amazon Q4 2025 earnings. amazon.science/blog/...-rufus
Amazon — COSMO	Commonsense knowledge graph built with LLMs, ACM SIGMOD 2024. amazon.science (COSMO, SIGMOD 2024)
Alibaba — URM	Jiang et al., “LLM as Universal Retriever in Industrial-Scale Recommender System” (2025). arxiv.org/abs/2502.03041
Alibaba — LUM & RecGPT	Yan et al., “Large User Model” (2025); Yi et al., “RecGPT Technical Report” (2025). arXiv 2025 (LUM); RecGPT report, Aug 2025
JD.com — GenCTR & GRAM	Kong et al., “Generative CTR Prediction...” (2025); JD “GRAM: Generative Retrieval and Alignment” (2025). arxiv.org/abs/2507.11246
User-class / cold-start	Netflix cold-start (Li et al., Nov 2025); video-platform cold-start study (Aug 2025); cold-start LLM survey (2025). arXiv 2025

Note: pre-GPT milestones (item-to-item CF 1998/2003; EGES & DIN 2018; JD Telepath 2017) are historical; all quantitative effectiveness figures above are from 2024–2026 publications.

Speaker notes

Consolidated citations for the appendix. All quantitative effectiveness numbers come from 2024–2026 sources; the only pre-2024 references describe historical milestones.